

Section 2

2025年12月9日 星期二 10:52

ADC Lab. 2
5th December, 2025.
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Reference: ADC Part 2, section 2.

Aim: know about and find the bandwidth limitation for LT1366. Use website to build a 4th order filter to filter 8kHz noise.

Procedure.

• Before the lab.

We look up datasheet for LT1366 and find its Gain-Bandwidth product (GBP) and transfer function.

FEATURES

- Input Common Mode Range Includes Both Rails
- Output Swings Rail-to-Rail
- Low Input Offset Voltage: 150µV
- High Common-Mode Rejection Ratio: 90dB
- High A_{OL} : >10V/V Driving 10k Load
- Low Input Bias Current: 50nA
- Wide Supply Range: 1.8V to ±15V
- Low Supply Current: 375µA per Amplifier
- High Output Drive: 50mA
- 400kHz Gain-Bandwidth Product
- Low Noise: 6.5nV/√Hz
- Stable for Capacitive Loads Up to 1000pF

Fig2.1

We found that.

Each amp has its limit in bandwidth it amplifies, when gain of amp circuit rises, bandwidth must drop.

Gain × bandwidth = constant, which is GBP.

So if we want to find GBP on graph.

We find gain = 1 (0dB), and the frequency of that point is GBP.

Physical meaning of this point is: $\frac{V_o}{V_i} = 1$, amp at this point cannot do any amplifying, it's right at its limitation and ability.

GBP = 400kHz.

• Gain Bandwidth Product.

Opamps have built-in low-pass filtering to ensure circuits remain stable.

To find GBP, we build following inverting amp.

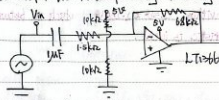


Fig2.3

∴ The circuit has a high-pass. (set $f_{c1} = R \Rightarrow 10k \Rightarrow 15.9k$)

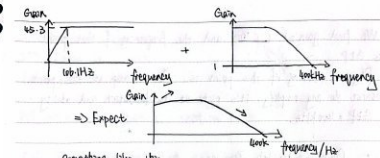
$$\frac{V_o}{V_i} = \frac{R_f}{R_i} = \frac{10k}{10k} = 1$$

$$A_{OL} = -45.3 \text{ V/V}$$

$$H_A = -45.3; \text{ with } f_c = \frac{1}{2\pi \times R \times C} = \frac{1}{2\pi \times 10k \times 10^{-8}} = 159.15 \text{ Hz}$$

And amp has a buffer in low-pass.

∴ Graph should look like.



Something like this.

And plot is as expected.

We have signal 50mV & 0V offset.

After amplifying, 50mV × (-45.3) = -2.265V.

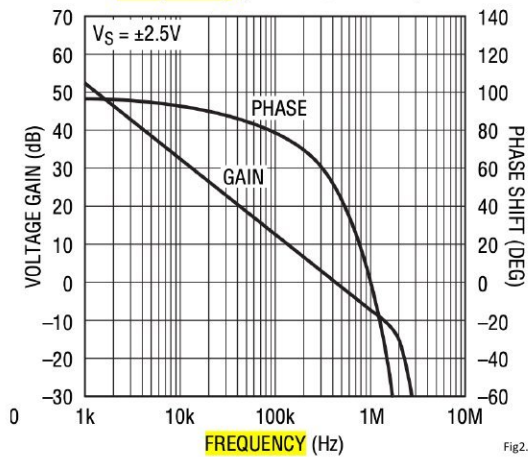
∴ Region is within -2.5V & 2.5V, which is between

0 ~ 5V. Characteristic is ideal.

f / Hz	Vout / V	Vin / V	Gain	Phase	dB
100	1.5	0.057	26.31579	-57	28.40433
320	2.24	0.056	40	-23.077	32.0412
600	2.36	0.0552	42.75362	-11.196	32.61946
1000	2.43	0.0544	44.66912	-3.579	33.00015
3200	2.32	0.056	41.42857	13.936	32.346
6000	2.1	0.0568	36.97183	26.024	31.35742
10000	1.74	0.0568	30.6338	41.04	29.72402
32000	0.74	0.0576	12.84722	-291.032	22.17618
50000	0.5	0.0584	8.561644	86.429	18.65114
60000	0.42	0.0584	7.191781	-269.783	17.13673
80000	0.34	0.0592	5.743243	-267.134	15.18314
100000	0.272	0.0632	4.303797	92.16	12.67704

Fig2.5

Gain and Phase Shift vs Frequency (LT1366/LT1367)

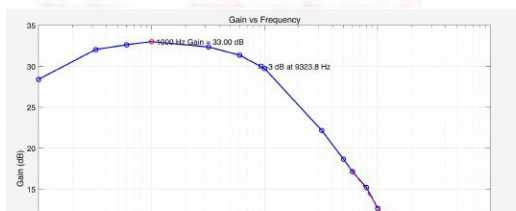
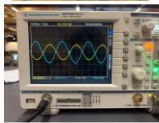


LT1366 TPC14

Fig2.2

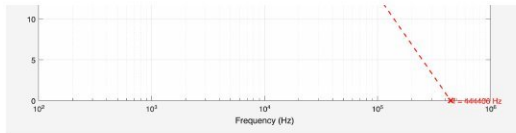


Fig2.4, 2.5



P252_fig

Fig2.6



We can use 2 ways to determine GBW = 400kHz.

- Extrapolate transfer function to 0 dB. $f = 444.406 \text{ kHz} \approx 400 \text{ kHz}$
- Definition: On graph, when amp at moderate f , gain = 30dB.
 $\therefore \text{Gain} = 10^{\frac{30}{20}} = 10^{1.5} = 31.6$
 Bandwidth = f at $30 \text{ dB} = 30 \text{ dB} \Rightarrow 900 \text{ Hz}$.
 $\therefore \text{GBW} = 10^{\frac{30}{20}} \times 900 = 402.015 \text{ kHz} \approx 400 \text{ kHz}$

We now consider phase:

For $G_{HP} = -\frac{R_2}{R_1} \frac{1}{1 + sRC}$ high-pass
 $L_A = -45.3 \text{ dB}$ $\rightarrow L_s = \frac{20}{s} \rightarrow 0.1 \text{ RC} \times \frac{1}{s} = 10 \text{ kHz}$
 $H_A = -40.3 \rightarrow L_s = 0 \rightarrow 10 \text{ RC} \times \frac{1}{s} = 1061 \text{ Hz}$

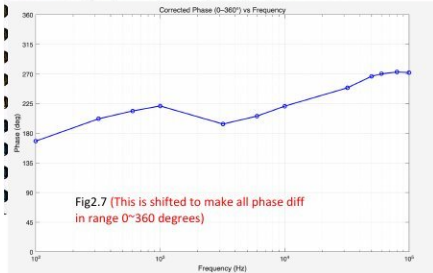
Phase plot shows a roll-off from 0 to -90 degrees at 1061 Hz.

Built-in low-pass phase:

From table. Also need to +180° as it's inverting.

For our phase-frequency:

$f < 100 \text{ Hz}$, $\phi = 180 + 90 - 180 = 90 - 90 \approx 0^\circ$
 $f > 100 \text{ Hz}$, $\phi = 180 - 180 \approx 0^\circ$
 where ϕ_0 is phase on datasheet.



Challenge: Active Filter

We use Analog Filter Wizard to build an active filter to filter 8000Hz noise.

to achieve, we need to make sure

- 200Hz & 2kHz within passband
- 8kHz within -40dB stopband (for gain).
- 4th order $\therefore$ only 2 poles
- 5V ~ 0V power supply.

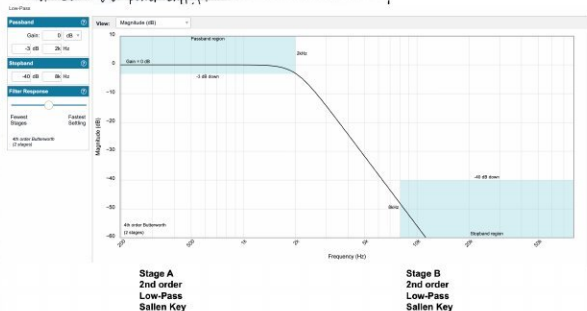


Fig2.8

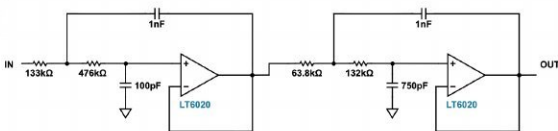
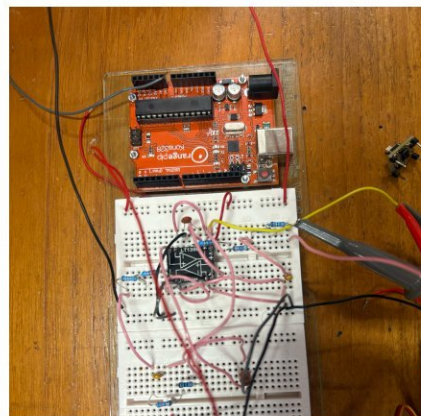


Fig2.9

It's a 4th order filter with 2 2nd order Sallen-Key filters

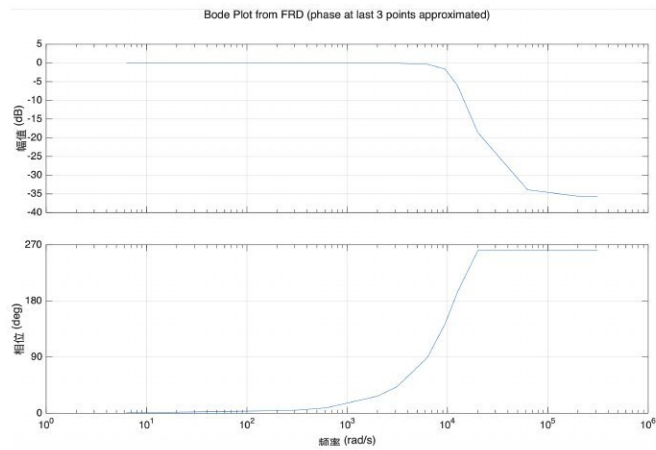
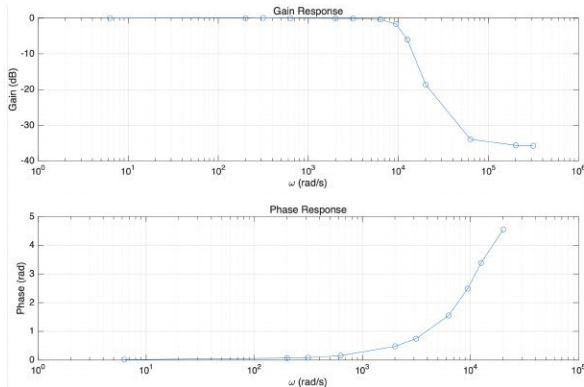
It's transfer function is like:

Frequency/Hz	Vin/V	Vout/V	Phase diff/degree	Gain	Gain/dB
1	0.984	0.992	0.781	1.0081301	0.0703315
32	1.02	1.02	3.873	1	0
50	1	1.01	4.773	1.01	0.0864275
100	1	1	8.813	1	0
320	0.992	0.984	27.375	0.9919355	-0.070331
500	0.984	0.976	42.603	0.9918699	-0.070906
1000	0.984	0.944	88.992	0.9593496	-0.360462
1500	0.984	0.816	142.707	0.8292683	-1.626099
2000	0.992	0.496	194.112	0.5	-6.0206
3200	0.992	0.116	260.896	0.1169355	-18.64107



10000	0.992	0.02 NaN	0.0201613	-33.90963
32000	0.968	0.016 NaN	0.0165289	-35.63511
50000	0.92	0.015 NaN	0.0163043	-35.75393

Fig2.10, 2.11, 2.12, 2.13



[Raw data.fig](#)

[Transfer f.fig](#)

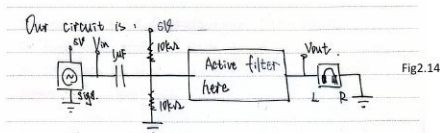


Fig2.14

Also capacitor & potential of 2.5V bias added for same reason.
I do a test on LTspice and it's working.

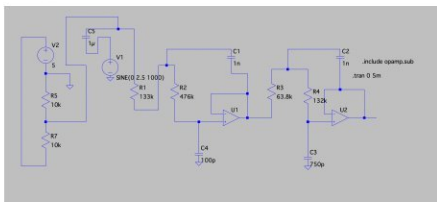
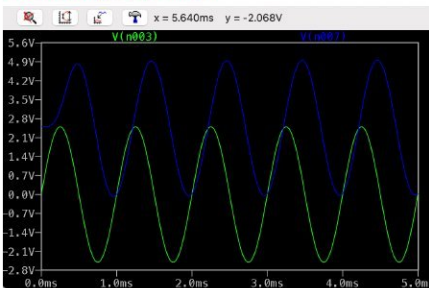


Fig2.15

At 1000Hz, it outputs amplitudes well:



At 8000Hz, it outputs nearly no amplitudes:

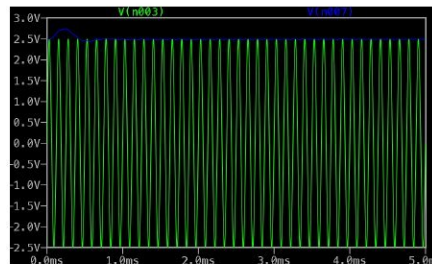


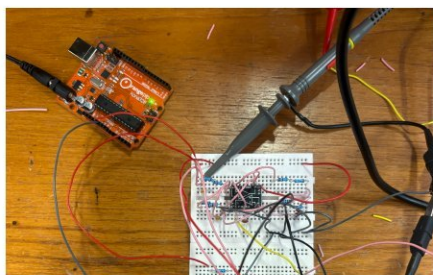
Fig2.16, 2.17

Sallen-key is a 2nd order filter, so 2 sallen-keys will bring something like

$$\text{Gain} = \frac{A}{B(j\omega)^2 + C(j\omega) + D} \cdot \frac{E}{F(j\omega)^2 + G(j\omega) + H}$$

which is 4th order

We connect channel and it really filter the noise.



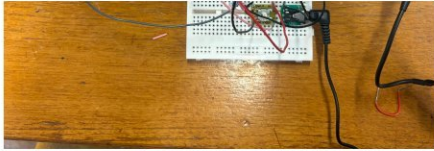
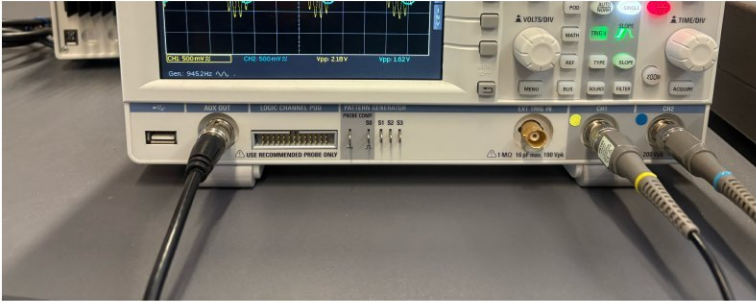


Fig2.18, 2.19



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